Original Article

Impact of Early Management in Osteoarticular Infections in Infants and Neonates: A Two-Year Retrospective Study

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Abstract

Background: Osteoarticular infections in neonates and infants, including osteomyelitis and septic arthritis, can cause irreversible joint damage, growth disturbance, deformity, and functional disability. Early recognition and definitive institutional treatment are therefore important. This study evaluated the clinical characteristics, risk factors, causative organisms, treatment, and relationship between delay in institutional presentation and outcome. **Methods:** This retrospective study included 40 eligible neonates and infants treated between June 2019 and May 2021. Data were obtained from the institutional Hosp Gestor database after Institutional Review Board approval and informed consent. Delay in institutional presentation was categorized as 0–7 days, 8–30 days, 1–3 months, and >3 months. Demographic characteristics, infection type and site, risk factors, laboratory findings, microbiology, treatment, and sequelae were analyzed. Statistical analysis was performed using SPSS 27.0 and GraphPad Prism 5; chi-square or Fisher's exact test was planned for categorical comparisons, with $P \leq 0.05$ considered statistically significant. **Results:** Of 40 patients, 8 (20%) were neonates (<28 days). Males constituted 70% of the cohort and the mean age was 5.4 months. Septic arthritis was the most common diagnosis (53%), followed by osteomyelitis (43%) and concomitant septic arthritis with osteomyelitis (4%). The hip was the most frequent site of septic arthritis (76.2%). Seventeen patients (42.5%) had a history of previous infection or trauma; pulmonary disease accounted for 9 of these (52.9%). Atypical associated factors included preterm delivery/low birth weight, congenital talipes equinovarus, intramuscular injection, scabies, and bacterial conjunctivitis. Eleven patients (27.5%) had received treatment elsewhere, though details of agents and duration were unavailable. *Staphylococcus aureus* was the commonest isolated organism (35%); *Acinetobacter haemolyticus* was isolated in 2.5%. Surgical cleaning/arthrotomy and drainage was the commonest procedure. Institutional delay was categorized as 0–7 days (19 patients), 8–30 days (10), 1–3 months (8), and >3 months (3). The most severe documented sequelae occurred with prolonged delay, including epiphyseal growth arrest, hip dysplasia, chronic osteomyelitis with recurrent pus formation, and gap non-union; category-wise complication counts were not available in the available dataset, so a valid inferential p-value for delay versus outcome could not be calculated. **Conclusion:** The most severe documented sequelae occurred among patients with prolonged delay in institutional presentation. Because category-wise complication counts were unavailable, the observed pattern should be interpreted descriptively rather than as a statistically confirmed association. The findings support prompt recognition and referral of neonatal and infant osteoarticular infections, particularly in settings where delayed presentation and treatment by informal health practitioners may occur.

Key Words: Infants, Neonates, Osteoarticular infection, *Staphylococcus aureus*, Time-to-treatment, Treatment outcome

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I. Introduction

Osteoarticular infections in infants and neonates, including osteomyelitis and septic arthritis, may progress rapidly and can result in systemic illness, irreversible joint damage, growth disturbance, limb deformity, and functional disability. Early treatment is therefore important to reduce permanent damage and sequelae[1-3].

In clinical practice, delayed institutional presentation may occur because of limited awareness, initial treatment outside formal healthcare facilities, referral delays, or failure to recognize early manifestations. Early radiographs may also be normal, making clinical assessment and appropriate use of ultrasonography or magnetic resonance imaging important when septic arthritis or osteomyelitis is suspected.

The present study evaluates the demographic characteristics, atypical presentations, risk factors, causative organisms, treatment patterns, and outcomes of osteoarticular infections in neonates and infants, with particular attention to the timing of institutional presentation.

Objectives

1. To identify demographic characteristics, atypical presentations, risk factors, and causative organisms in neonatal and infant osteoarticular infections.
2. To assess the role of definitive institutional management and document associated sequelae.
3. To compare clinical outcomes across categories of delay in institutional presentation.

II. Materials and Methods

Study design and setting

This was a retrospective observational study conducted after Institutional Review Board approval. Patients were identified from the institutional Hosp Gestor database. The study period was June 2019 to May 2021.

Participants

Forty eligible patients were included. Six additional patients were excluded because of incomplete charts. The 40 included patients were not a consecutive institutional series, as case selection depended on the available eligible records. Neonates were defined as patients younger than 28 days.

Inclusion and exclusion criteria

Neonates and infants admitted during the study period with a clinical and laboratory diagnosis of osteoarticular infection were eligible. Patients with incomplete charts were excluded.

Variables and definitions

Variables included age, sex, type and anatomical location of infection, delay in institutional presentation, previous infection or trauma, comorbidities and other reported risk factors, laboratory findings, microbiological results, treatment, complications, and sequelae. Delay in institutional presentation was categorized as 0–7 days, 8–30 days, 1–3 months, and >3 months.

Sample size

The sample size was calculated using Epi Info version 3.5.3, resulting in a target sample size of 40 patients.

Statistical analysis

Data were entered into Microsoft Excel and analyzed using SPSS version 27.0 and GraphPad Prism version 5. Continuous variables were summarized using mean and standard deviation; categorical variables were summarized using counts and percentages. Chi-square or Fisher's exact test was planned for comparison of categorical variables, as appropriate, with $P \leq 0.05$ considered statistically significant. Because the number of complicated and uncomplicated patients within each delay category was not available, an inferential comparison of delay categories with complication status could not be performed.

III. Results

The final study sample comprised 40 patients. Eight (20%) were neonates younger than 28 days. Males predominated (70%), and the mean age was 5.4 months.

Diagnosis	n	%
Septic arthritis	21	52.5
Osteomyelitis	17	42.5
Septic arthritis + osteomyelitis	2	5.0
Total	40	100

Table 1: *Types of osteoarticular infection (n=40)*

Septic arthritis was the most common diagnosis. The hip was the most frequent site of septic arthritis, involving 76.2% of affected patients. Osteomyelitis most commonly involved the tibia.

Risk factors and previous infections

Seventeen patients (42.5%) had a history of previous infection or trauma. Pulmonary diseases, including bronchiolitis and pneumonia, were the most frequent previous infections, accounting for 57.14% (n=9) of these histories. Atypical associated factors included preterm delivery/low birth weight (20%), congenital talipes equinovarus (5%), previous intramuscular injection (5%), scabies (5%), and bacterial conjunctivitis (5%).

Treatment before institutional presentation

Twenty-nine patients (72.5%) had not received treatment before their first presentation. Eleven patients (27.5%) had received some form of treatment outside the institution before presentation. The specific antimicrobial agents, route, and duration of treatment were not available in the records supplied for analysis; the effect of prior antimicrobial exposure on culture yield could therefore not be reliably assessed.

Microbiology and laboratory findings

Staphylococcus aureus was the most commonly isolated organism (35%). *Acinetobacter haemolyticus* was identified as an atypical organism in 2.5% of cases. CRP and ESR were reported to be elevated in the study population.

Delay in institutional presentation and outcome

The available clinical records demonstrated progressively more severe sequelae with prolonged delay. Documented complications included epiphyseal growth arrest, dysplasia of the hip, chronic osteomyelitis with recurrent pus formation, and gap non-union. The most severe outcomes were observed among patients presenting after prolonged delay, particularly those with delay exceeding 3 months. Because the supplied data do not specify the number of complicated versus uncomplicated patients within each delay category, an exact chi-square/Fisher's exact p-value cannot be reported without introducing unsupported assumptions.

Delay category	n	%
0–7 days	19	47.5
8–30 days	10	25.0
1–3 months	8	20.0
>3 months	3	7.5
Total	40	100

Table 2: *Delay in institutional presentation (n=40)*

Treatment

Method	Description
Surgical cleaning/incision & drainage	Commonest operative approach
Arthrotomy and drainage	Common, particularly for septic arthritis
Intravenous antibiotics	Used as part of institutional treatment
Saucerization/sequestrectomy	Used where indicated
Debridement	Used where indicated
Osteotomy	Used for selected residual deformity

Table 3: *Treatment methods reported in the study*

IV. Discussion

The present study highlights the clinical importance of timely institutional management of osteoarticular infections in neonates and infants. Septic arthritis was the predominant infection, and the hip was the most frequently involved joint. The predominance of *Staphylococcus aureus* is consistent with the established microbiology of paediatric osteoarticular infection[4-9].

The study also identified potentially relevant preceding conditions, including pulmonary infections, preterm delivery and low birth weight, congenital talipes equinovarus, intramuscular injection, scabies, and bacterial conjunctivitis. These findings may help clinicians maintain a high index of suspicion in infants with fever, irritability, limb non-use, swelling, or reduced joint movement.

Delayed presentation in this study population was associated descriptively with more severe sequelae. Documented reasons for delay included lack of awareness and inappropriate treatment by informal health practitioners, which may contribute to delayed referral and prolonged infection before definitive treatment. In settings with limited access to specialist care, early recognition and referral are therefore particularly important.

Kocher et al. described clinical approaches to diagnosis and management of septic arthritis of the hip, while other studies have emphasized early antimicrobial therapy and appropriate surgical drainage when indicated[7-9]. Empirical antimicrobial therapy is generally initiated after appropriate cultures have been obtained when clinically feasible, with subsequent adjustment according to microbiological results and clinical response.

Early radiographs may not demonstrate osteoarticular infection, particularly in the initial stages. Ultrasonography is useful for detecting joint effusion and guiding aspiration in suspected septic arthritis, especially at the hip. Magnetic resonance imaging is more sensitive for early marrow, metaphyseal, epiphyseal, and soft-tissue involvement and can be considered when the diagnosis remains uncertain or osteomyelitis is suspected despite non-diagnostic initial radiographs.

The study's observation of severe sequelae after prolonged delay is clinically important, because damage to the growing epiphysis and physis may produce growth arrest, deformity, dysplasia, limb-length discrepancy, and persistent functional impairment even after eradication of infection.

Limitations

This study has several limitations. The sample size was small (40 patients), the study was conducted at a single institution, and the included cases were not a consecutive series, introducing the possibility of selection bias. The study was retrospective and depended on information available in the Hosp Gestor database; six patients with incomplete charts were excluded. Details of treatment received outside the institution, including antibiotic type and duration, were unavailable, limiting assessment of its effect on culture yield and outcomes. Reasons for delayed presentation were not systematically quantified. Long-term follow-up beyond the documented sequelae was limited. Finally, the supplied aggregate data did not provide category-wise numbers of complicated and uncomplicated patients, preventing calculation of a valid inferential p-value for the association between delay category and outcome.

V. Conclusion

Osteoarticular infections in neonates and infants can result in serious and irreversible sequelae when recognition and definitive treatment are delayed. This study demonstrates that prolonged delay in institutional presentation was accompanied by more severe documented complications, including epiphyseal growth arrest, hip dysplasia, chronic osteomyelitis with recurrent pus formation, and gap non-union. Lack of awareness and inappropriate treatment by informal health practitioners were identified as important contributors to delay. Early clinical suspicion, appropriate imaging and microbiological evaluation, prompt referral, and timely definitive institutional management are essential to improve outcomes.

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